AI Study Notes

1. Artificial Intelligence

Artificial Intelligence (AI) is a broad field of computing focused on building systems that can perform tasks that normally require human intelligence.

2. Machine Learning

Machine Learning (ML) is a branch of AI in which systems learn patterns from data and use those patterns to make predictions or decisions.

3. Deep Learning

Deep Learning is a part of machine learning that uses neural networks with multiple layers to learn complex patterns from data.

4. Types of Machine Learning

Common types include supervised learning, unsupervised learning, and reinforcement learning.

5. Neural Networks

Neural networks are computational models made of connected units that transform input data through layers and learn useful representations.

6. Generative AI

Generative AI refers to AI systems that can create new content such as text, images, audio, or code.

7. Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with text generation. Documents are processed into smaller chunks, converted into embeddings, and relevant chunks are retrieved for a user question.

8. Embeddings

Embeddings are numerical vector representations of text or other data. Text with similar meaning can have similar vector representations.

9. Vector Databases and FAISS

FAISS (Facebook AI Similarity Search) is a library designed for efficient similarity search over dense vectors. In a RAG system, document chunks can be converted into embeddings and stored in a FAISS index.

10. Key Takeaway

AI includes Machine Learning and Deep Learning. Generative AI creates new content, while LLMs enable language-based applications. Embeddings, vector indexes, and RAG help build AI systems that work with domain-specific information.